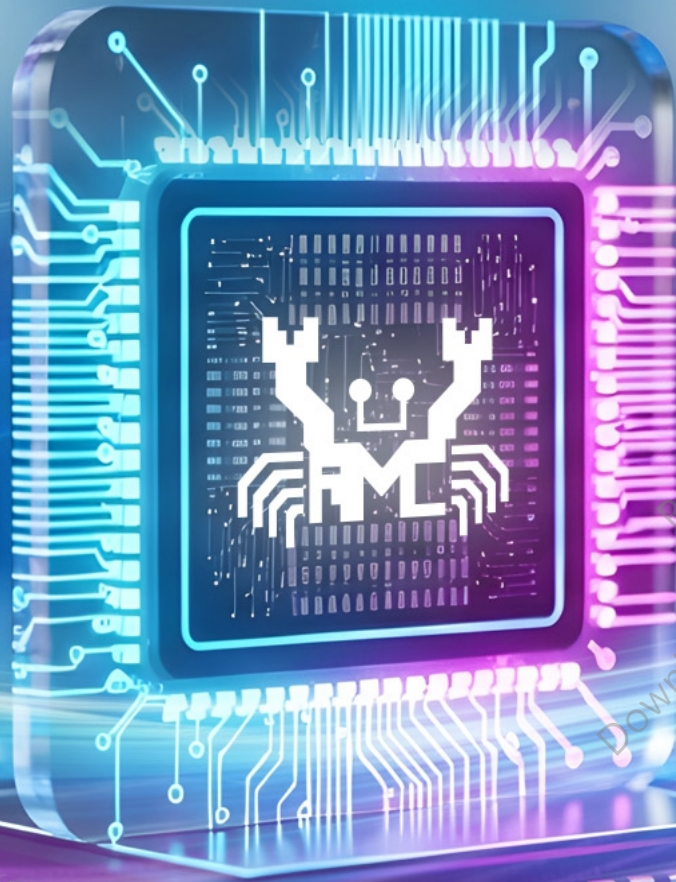


vhal client  
changxi\_huang  
2026/01/29



**Realtek**  
Powering the AI World

Connected Media  
Realtek Semiconductor Corporation  
Q1 2026

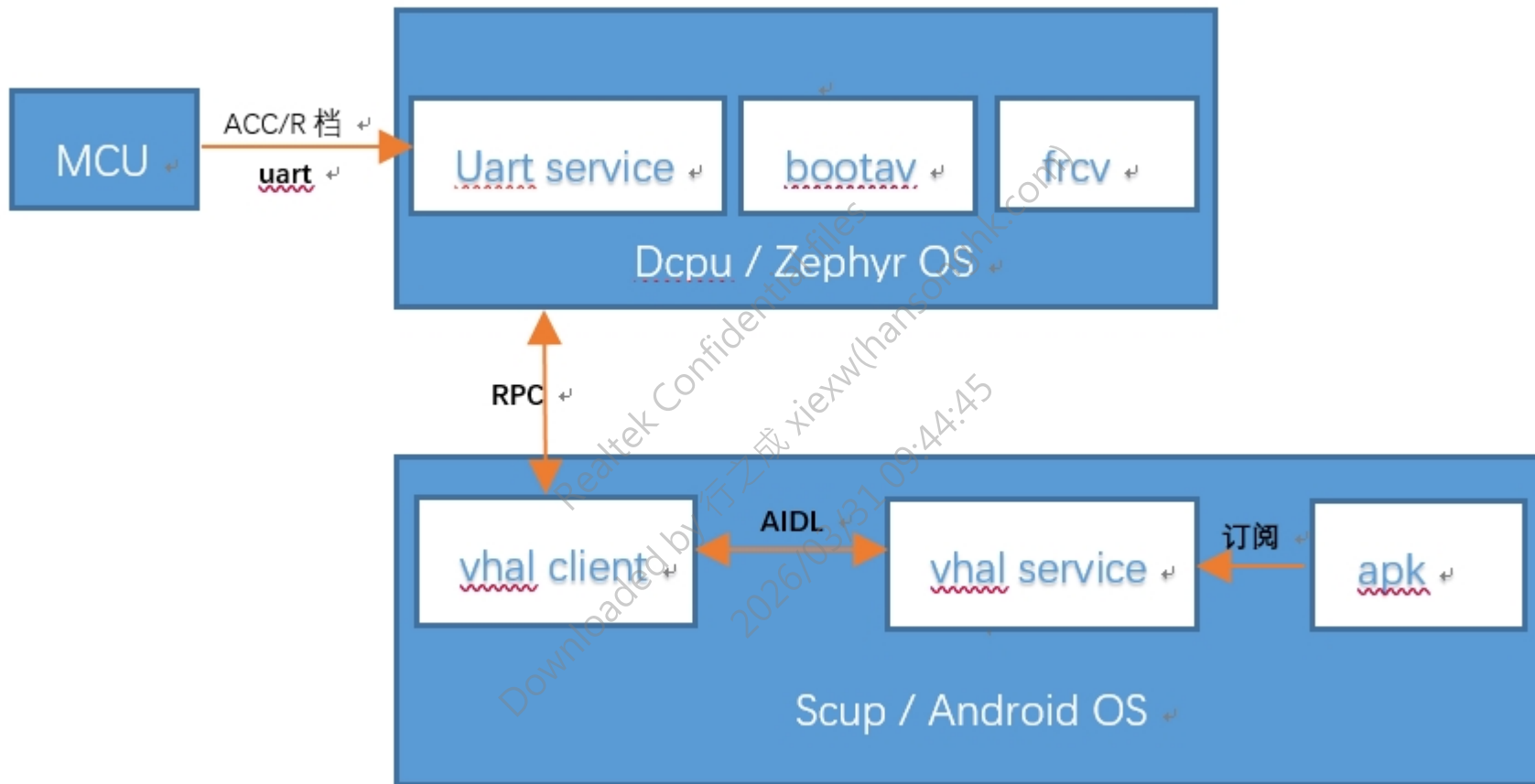
Downloaded by 行之成 xiexw(hansonghk.com)  
2026/03/31 09:44:45

# Overview

- Block Diagram
- 重点流程及客制化
- 已对接实现的RPC
- 订阅的属性
- 常见的异常案例分析

Realtek Confidential files  
Downloaded by 行之成 xiexw(hansonghk.com)  
2026/03/31 09:44:45

# Block Diagram



[VHAL Manual - DHC\\_Automotive - WIKI](#)



# RPC处理流程及客制化

InitKR4RPCProxy//初始化 ↵

prepareCLNT//参数配置 ↵

vhal\_RegisterVhalToSystemPrivateInfoCallback//注册 callback ↵

Callback 有 rpc msg? ↵

处理收到的 rpc cmd ↵

```
static HRESULT*
parseSrvCmdAllInfo(VHAL_RPC_KR4_PRIVATEINFO_TO_SYS *p_param,
RPC_STRUCT *rpc, HRESULT *retval){
    switch (p_param->type){
        case ENUM_PRIVATEINFO_TO_SYS_POWER_ACC:
            //add your code here
            ....
            break;
        case ENUM_PRIVATEINFO_TO_SYS_POWER_PANEL_OFF:
            ....
            break;
        case....
    }
}
```

客制化时主要就是增加或者删减RPC,按需处理(  
比如目前没有想要的消息，可以新增RPC)

# vhal属性处理流程及客制化

```
typedef struct {  
    int propertyId; //vahl 属性ID  
    void (*rpcFunction)(const std::vector<int32_t>&); //该ID变化时的处理函数  
} Subscribe;
```

1) 定义想要处理的属性ID(需要在vhal service中注册)

```
static const Subscribe propertyId_subscribe[] = {  
    {557843458, onCameraReady},  
    {toInt(VehicleProperty::AP_POWER_STATE_REPORT), onPowerStateChanged},  
    {toInt(VehicleProperty::CLUSTER_REQUEST_DISPLAY), onClusterRequestDisplay},  
    {0x21400403, onFAVMStateChanged},  
};
```

2) 监听订阅上述定义的vhal 属性变化

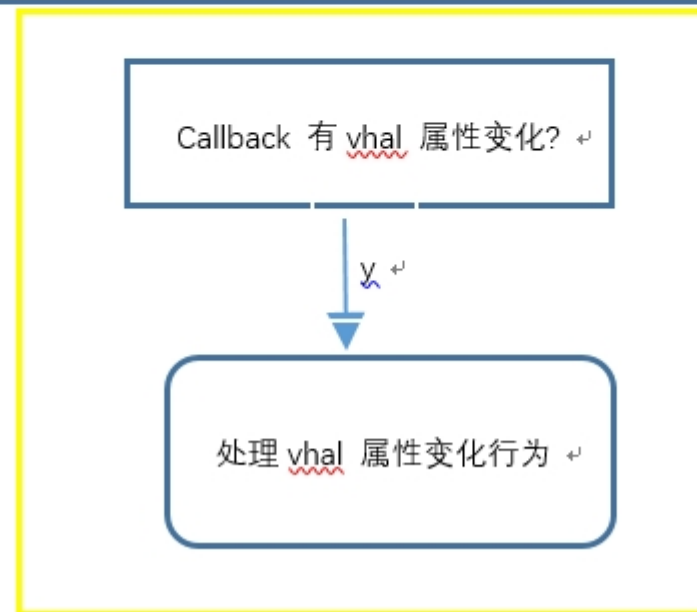
```
for(const auto& item : propertyId_subscribe){  
    int32_t res = 0;  
    auto status = g_VhalClient->subscribeToVHal(item.propertyId, &res);  
    if(!status.isOk()){  
        ALOGE("%s:%d subscribe %d failed", __func__, __LINE__, item.propertyId);  
    }  
}
```

3) 557843458属性变化的处理函数(这里是调用rpc api发送消息给DCPU)

```
void onCameraReady([[maybe_unused]] const std::vector<int32_t>& value)  
{  
    VHAL_RPC_KR4_PRIVATEINFO_PARAMETER param;  
    memset(&param, 0x0, sizeof(param));  
    param.type = ENUM_PRIVATEINFO_POWER_SCPU_AOSP_RCV_RDY;  
    ALOGI("%s:%d send rpc cmd, waiting for response", __func__, __LINE__);  
    sendMsg(&param);  
}
```

上述 `propertyId_subscribe[]` 数组可以按需自行增减，处理函数同样按自己需求修改。

```
getPropertyChangeListener() // 获取监听对象  
registerCallBack() // 注册回调函数-用于处理 vhal service 的通知  
subscribeToVHal() // 订阅 vhal 属性
```



# 已对接实现的RPC

ENUM\_PRIVATEINFO\_TO\_SYS\_VHAL\_SERVICE\_CMDALL\_SEND //cluster定时推送速度 挡位等信息  
ENUM\_PRIVATEINFO\_TO\_SYS\_POWER\_PANEL\_ON/OFF //开关屏幕  
ENUM\_PRIVATEINFO\_TO\_SYS\_POWER\_SUSPEND  
--SUSPEND\_DATA0\_prepare //准备进S3 VehicleApPowerStateReq::SHUTDOWN\_PREPARE  
--SUSPEND\_DATA0\_cancel //取消进S3 VehicleApPowerStateReq::CANCEL\_SHUTDOWN  
--SUSPEND\_DATA0\_prepare\_done //准备完成 VehicleApPowerStateReq::FINISHED  
--SUSPEND\_DATA0\_prepare\_ack //Android进入S3时通知cluster  
ENUM\_PRIVATEINFO\_TO\_SYS\_POWER\_ENTER\_NORMAL //”换手”  
ENUM\_PRIVATEINFO\_TO\_SYS\_POWER\_ACC //点火/熄火  
ENUM\_PRIVATEINFO\_TO\_SYS\_MISC\_REAR\_GEAR //进出R档  
ENUM\_PRIVATEINFO\_TO\_SYS\_MISC\_UPDATE\_CUSTOMIZE\_RAW  
ENUM\_PRIVATEINFO\_MCU\_SERVICE\_INIT //初始化兼回传MCU连接状态  
ENUM\_PRIVATEINFO\_POWER\_SCPU\_INIT //往DCPU发送的第一条msg  
ENUM\_PRIVATEINFO\_POWER\_SCPU\_AOSP\_RCV\_RDY //camera ready  
ENUM\_PRIVATEINFO\_POWER\_SCPU\_ANDROID\_BOOT\_FINISH  
ENUM\_PRIVATEINFO\_MCU\_SERVICE\_STOP\_BOOTAV  
ENUM\_PRIVATEINFO\_MCU\_SERVICE\_AAOS\_CLUSTER //投屏  
ENUM\_PRIVATEINFO\_MCU\_SERVICE\_CLUSTER\_VER //cluster版本信息

# 订阅的属性

| Property id                            | 描述                |
|----------------------------------------|-------------------|
| 557843458                              | camera ready      |
| 557843472                              | VENDOR_BOOT_STATE |
| 557843465                              | ACC_STATE         |
| 557843464                              | EXIT_REVERSE      |
| 554696961                              | CLUSTER_VERSION   |
| VehicleProperty::HW_KEY_INPUT          | Acc点火开关屏          |
| VehicleProperty::PERF_VEHICLE_SPEED    | 速度                |
| VehicleProperty::GEAR_SELECTION        | 挡位                |
| VehicleProperty::AP_POWER_STATE_REQ    | CPMS状态控制          |
| VehicleProperty::AP_POWER_STATE_REPORT | CPMS状态            |
| VehicleProperty::HW_CUSTOM_INPUT       | 按键开关屏幕            |

# 常见的异常案例分析

vhal client service在logcat中的关键字是: **ClusterRpc**

没有收到进出R档信息

---首先确保MCU连接正常; 其次确保camera apk 发出camera ready

vendor.rtk.car.boot.state 属性没有正常设1

--- 和上面处理一样

log总是显示MCU未连接，特别是无MCU的场景

---vhal client service最开始没考虑无MCU的场景，后来才加上；请升级vhal client service

launcher界面提示"当前处于驾驶模式,apk不可用"

---MCU发送挡位信息错误；请客户正常对接MCU，非驾驶状态时一定要不要发D档

acc on/off未正确响应

---如果是公版模式，目前该流程还存在bug；如果是客户对接，请确保mcu/cluster端按规范对接acc 流程，抓取logcat 以及 /mnt/vendor\_logs/LastLogs/下日志文件分析





Thank You

***REALTEK AI: Changing the Future***